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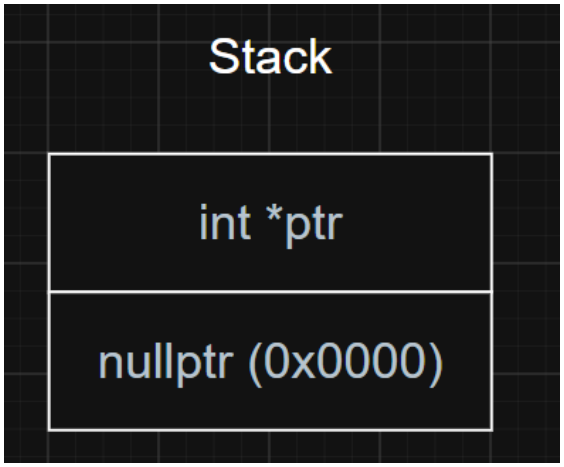
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|---|---|---|---|
| **COURSE CODE/TITLE:** | CPE 010 - Data Structures and Algorithms | **SECTION:** | CPE21S2 |
| **DATE PERFORMED:** | Jul 15, 2026 | **DATE SUBMITTED:** | Jul 15, 2026 |
| **NAME:** | Antonio, Christine Mae P. | **INSTRUCTOR:** | Engr. Jimlord M. Quejado |

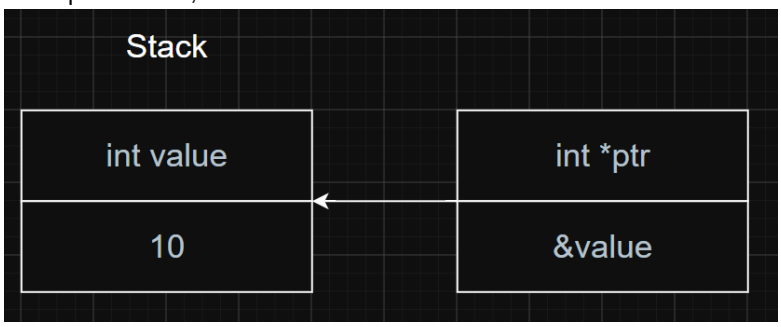
**SEATWORK NO. 2.1** **Memory Diagram Exercise**

**1. PROCEDURE OUTPUT**

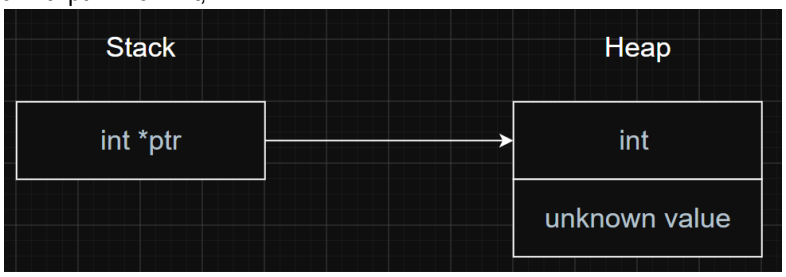
1. int *ptr = nullptr;



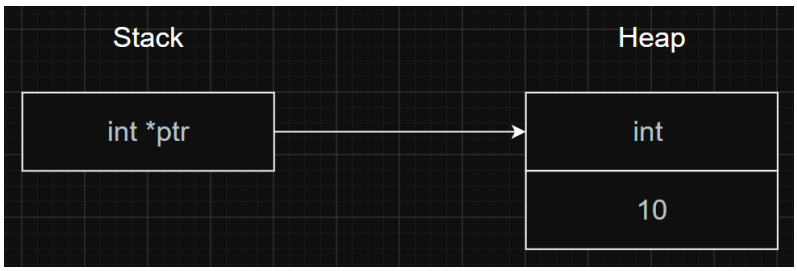

2. int value = 10;
   int *ptr= &value;



3. int *ptr = new int;




4. int *ptr = new int(10);




5. Differentiate Constructor, Destructor, Copy Constructor, Copy Assignment Operator.

**Constructor** - initilizes the object of the class
**Destructor** – it is automatically called when an object is destroyed.
**Copy Constructor** - creates a new object as a copy of another object.
**Copy Assignment Operator** - copies one existing object into another existing object.

## 2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

## 3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

**Tools Used:** **Claude**

*Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.*

**Purpose of Use:** **Example generation**

*Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.*

**Extent of Use:** **The AI was used as a guide on how to make a memory diagram.**

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*Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.*

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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